SERANGOON JUNIOR COLLEGE
General Certificate of Education Advanced Level
Higher 1

CANDIDATE NAME

CLASS

INDEX NUMBER

CHEMISTRY **8872/02**
Preliminary Examination **22 August 2013**
Paper 2 **2 hours**

Additional Materials: Data Booklet, Cover Page for Section B
Writing Papers, Graph Paper

READ THESE INSTRUCTIONS FIRST

Write your name and class on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams, graphs or rough work.

SECTION A:
Answer **all** questions in the space provided in the booklet.

SECTION B:
Answer any **two** questions on separate answer papers.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [ ] at the end of each question or part question.

For Examiner's Use				
MCQ P1				
Section A	1	2	3	Total
B4				
B5				
B6				
Total				/80

This document consist of **14** printed pages and **2** blank pages

Section A

Answer **all** the questions in this section in the spaces provided.

For Examiner's use

1 **(a)** Arsenic acid, H_3AsO_4, is a colourless acid that is used as a wood preservative and as a reagent in the synthesis of dyestuff. It is slowly formed when elemental arsenic, As, is treated with ozone in the presence of moisture. Oxygen gas is known to be a side product for the reaction.

(i) Write a balanced chemical equation for the above mentioned treatment process.

....

(ii) Arsenic acid can oxidise iodide ions to iodine and itself is reduced to H_3AsO_3.
Write a half-equation for the reduction of arsenic acid to H_3AsO_3.
Hence write an overall equation for the reaction between iodide ions and arsenic acid.

....

....

....

(iii) The iodine, I_2, that is liberated from the reaction between arsenic acid and iodide ions can be estimated by titration against a standard thiosulfate, $S_2O_3^{2-}$, solution.

$$I_2 + 2S_2O_3^{2-} \rightarrow 2I^- + S_4O_6^{2-}$$

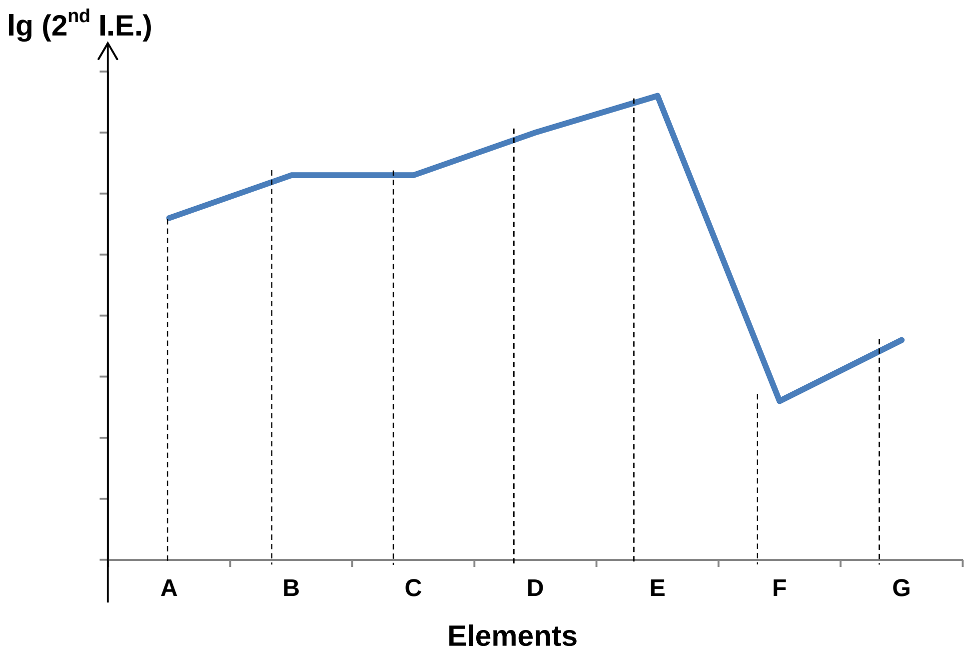
When a sample of 25.0 cm^3 of arsenic acid reacts with iodide ions, it was found that 23.30 cm^3 of 0.200 mol dm^{-3} solution of thiosulfate ions was required to react completely with the iodine liberated.

Calculate the concentration of the sample of arsenic acid.

[5]

For Examiner's use

(c) The following graph shows some data on consecutive elements from period 2 and period 3.



F is known to react with **C** to form an ionic compound.
G is a better conductor of electricity as compared to **F**.
The chloride of **F** gives a slightly acidic solution when dissolved in water.

(i) From the information given, identify **F**.
Hence, describe the type of bonding in **F** either in words or with a labelled diagram.

(ii) Suggest if the compound formed from **F** and **C** is soluble in water. You must support your answer with relevant explanations to gain full credit of this question.

....

....

....

[4]
Total: [9]

For Examiner's use

2 **(a)** For the reaction,

$$CH_3CH_2CO_2H\ (l) + CH_3OH\ (l) \rightleftharpoons CH_3CH_2CO_2CH_3\ (l) + H_2O\ (l)$$

The value of the equilibrium constant is 5.0.

(i) Suggest one way to increase the yield of the above reaction.

....

....

(ii) Explain, in terms of chemical bonding and structure, why the product, $CH_3CH_2CO_2CH_3$ has low solubility in water.

....

....

....

....

(iii) State the numerical value of the equilibrium constant for the reverse reaction.

....

(iv) Suggest the reagent and condition for the reverse reaction.

....

(v) In an experiment, 1 mol of $CH_3CH_2CO_2CH_3$ and 1 mol of water are mixed in V cm^3. Calculate the amount of acid present at equilibrium.

For Examiner's use

(vi) Suggest a chemical test to distinguish between $CH_3CH_2CO_2CH_3$ and $CH_3CH_2CO_2CH_2CH_3$.

....

....

....

....

....

Total: [8]

For Examiner's use

3 Glycolic acid (pK_a = 3.83), $C_2H_4O_3$, is a colourless and odourless crystalline solid that is highly soluble in water. It is the smallest α–hydroxy acid (AHA) and is commonly used in skin care products and most often as a chemical peel performed by plastic surgeons due to its ability to penetrate the skin effectively.

It can be prepared by the reaction of chloroethanoic acid (pK_a = 2.85) with hot sodium hydroxide followed by re-acidification.

(a) **(i)** Write a chemical equation for the net reaction that has occurred showing the displayed formulae of all organic compounds.

(ii) Explain why chloroethanoic acid has a lower pK_a as compared to glycolic acid.

....

....

....

....

....

....

(iii) Chloroethanoic acid is defined as a *weak acid*.
With the aid of an equation, explain the words in *italics*.

....

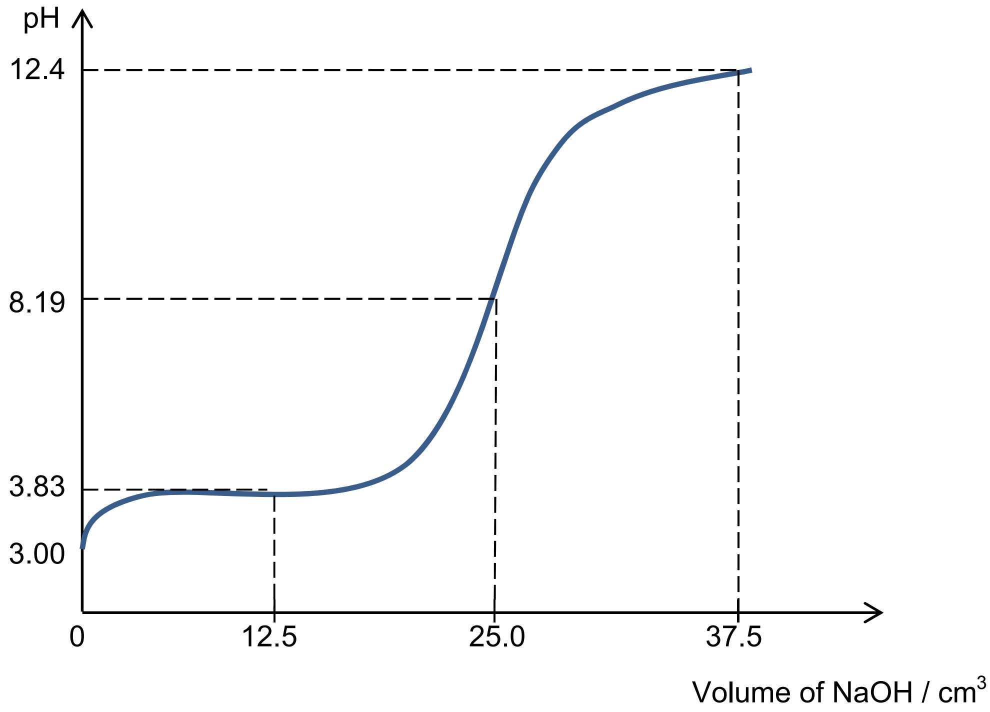
....

....

For Examiner's use

(iv) Besides pK_a, percentage dissociation is another way to measure the strength of an acid. Percentage dissociation is the ratio of the concentration of hydrogen ions to the concentration of the acid expressed in percentage.

An analysis was carried out on a 25 cm^3 sample of a skin care product containing glycolic acid by titrating with 0.125 mol dm^{-3} of sodium hydroxide. The graph obtained is shown below. With this graph, determine the percentage dissociation of glycolic acid.



(v) Prove that the final pH is 12.4 after 37.5 cm^3 of sodium hydroxide is added.

For Examiner's use

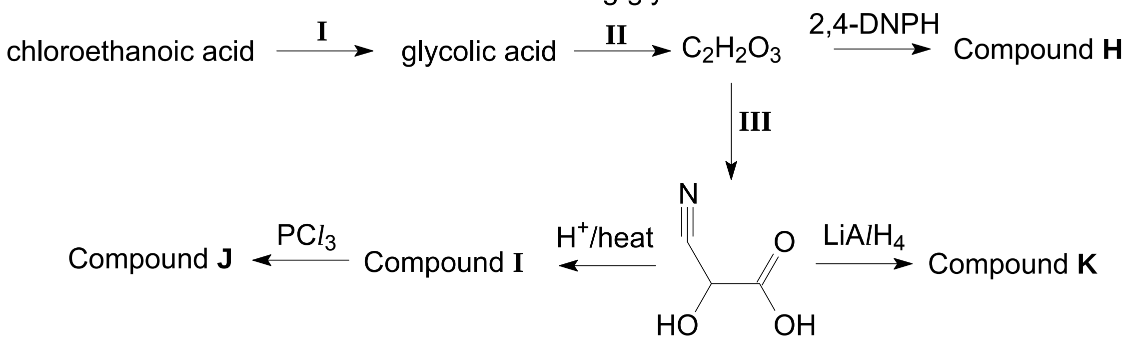
(vi) On the graph in **(a)(iv)**, indicate the region where there is a buffer solution. State the species involved in this buffer system.

(vii) Using an equation only, suggest how the pH remains fairly constant when small amount of sodium hydroxide is added to the solution in **(a)(vi)**.

....

[13]

(b) The reaction scheme shows reactions involving glycolic acid.



(i) State the reagents and conditions for reactions **II** and **III**.

Reaction **II**: ....
.
Reaction **III**: ....

(ii) Draw the structural formulae for compounds **H**, **I**, **J** and **K**.

H	**I**	**J**	**K**

For Examiner's use

(iii) With the aid of the *Data Booklet*, suggest what will happen to the rate of reaction for reaction **I** when chloroethanoic acid is replaced with bromoethanoic acid.

....

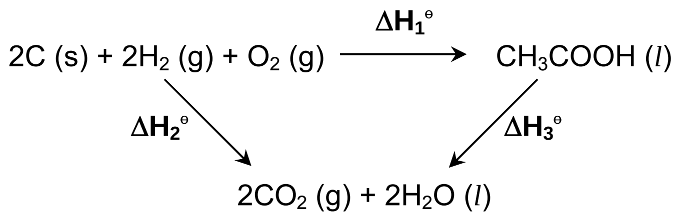
....

....

....

[8]

(c)



Using the following data and the energy cycle above, calculate the standard enthalpy change of combustion of ethanoic acid.

$\Delta H_c^\ominus$ carbon = -393 kJ mol^{-1}
$\Delta H_c^\ominus$ hydrogen = -286 kJ mol^{-1}
$\Delta H_f^\ominus$ ethanoic acid = -487 kJ mol^{-1}

[2]
Total: [23]

Section B

Answer two of the three questions in this section on separate paper.

4 (a) Using sodium, phosphorous and sulfur, describe the reactions of their oxides with either sodium hydroxide or hydrochloric acid or both.

[4]

(b) Propanone undergoes keto-enol tautomerism when reacted with iodine under an acidic medium at room temperature. The overall equation for this reaction is represented below:

$$CH_3COCH_3 + I_2 \xrightarrow{H^+} CH_3COCH_2I + HI$$

To investigate how the rate of reactions depends on the concentration of each of the three reactants, two experiments were carried out. In each experiment the concentrations of two reactants were in excess and kept constant, whilst the concentration of the third reactant was measured over time. It is known that the order of reaction with respect to hydrogen ion is first order.

	reaction 1	reaction 2
time / s	$[CH_3COCH_3]$ / mol dm^{-3}	$[I_2]$ / mol dm^{-3}
0	1.50	1.50
30	0.96	1.28
60	0.62	1.05
90	0.38	0.83
120	0.25	0.60
150	0.15	0.38
180	0.09	0.15

(i) Plot a graph of these results, putting all of the data on the same axes. Label each curve clearly.

(ii) Use your graph to determine the order of reaction for each of the two reactants. Justify your answer in each case.

(iii) Use your answer from **(b)(ii)** to write a rate equation for the reaction.

(iv) Explain how the rate of reaction would change if chlorine is used instead of iodine.

(v) With an aid of a suitable diagram, explain the effect on the rate of reaction if it is carried out in an ice bath.

(vi) Propanone can also react with iodine under basic medium resulting in a different organic product. Write an equation for this reaction, indicating the type of reaction involved and the observation.

[14]

(d) Propose a chemical test to differentiate CH_3COCH_3 from CH_3COCH_2I.

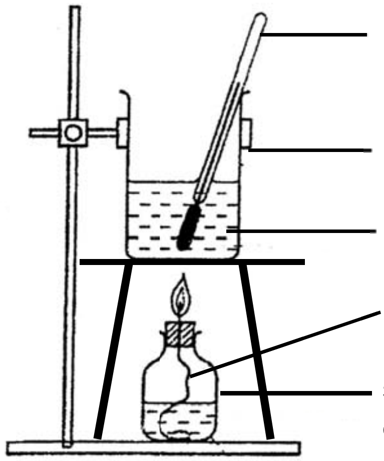
[2]
Total: [20]

5 **(a)** The table below shows the sources and the enthalpy changes of combustion of two common fuels used in vehicles.

Fuel	Main Sources	Enthalpy change of combustion / kJ mol^{-1}
Octane	Refined from crude oil	– 5460.6
Ethanol	Fermentation	–1359.8

(i) Define standard enthalpy change of combustion using ethanol as an example.

(ii) The figure below shows a diagram of a calorimeter used by a student to determine the enthalpy change of combustion of ethanol.



When 0.960 g of ethanol was combusted, the energy evolved heated 100 cm^3 of water from 25.0 °C to 72.6 °C.

Using the information provided, determine the percentage efficiency, x, of this experiment, leaving your answer to three significant figures.

(iii) The fuel value (in kJ g^{-1}) of a substance is the heat energy released when 1 g of the substance is combusted.

Calculate the fuel value for each of the two fuels.

Hence, suggest a reason why octane has an advantage over ethanol as fuel for vehicles.

(iv) Name the isomer of octane which has the lowest boiling point.

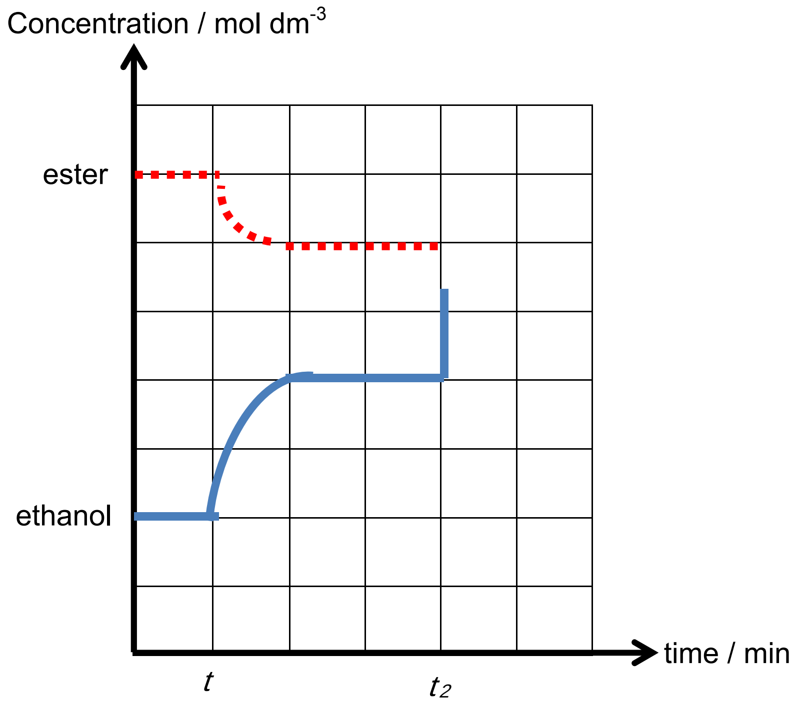
[7]

(b) Ethanol has many uses in the organic laboratory. One of its common uses is to manufacture esters.

$$2CH_3CH_2OH + HOOCCH_2COOH \rightleftharpoons CH_3CH_2OCOCH_2COOCH_2CH_3 + 2H_2O$$

(i) Deduce the effect on the equilibrium constant if a catalyst is introduced to the reaction chamber.

(ii) Equal amounts of ethanol and propanedioic acid reacted together to reach equilibrium.



Upon establishing equilibrium at time t, temperature of the reaction is decreased. Determine whether the forward reaction is exothermic or endothermic.

(iii) State the observed changes in concentration of ethanol and ester as a result of the change that occurred at t_2.

(iv) A student intern at the laboratory made the following claim:

For the esterification reaction above, the enthalpy change is known as the enthalpy change of neutralisation.

However, the enthalpy change cannot be calculated accurately by using only the bond energies in the data booklet.

Comment on the intern's claim.

[6]

(c) An ester **L**, $C_8H_{13}O_2Cl$ undergoes reaction to give ethanol and acid **M** $C_6H_9O_2Cl$.

Decolourisation occurs when 1 mole of ester **L** reacts with 1 mole of liquid bromine.

When acid **M** reacts with hot acidified potassium manganate (VII), compound **N** is formed together with 2 moles of gas. Compound **N** is inert towards sodium carbonate powder and Fehling's reagent but produces orange crystals when reacted with hydrazine.

1 mole of compound **N** can react with alkaline aqueous iodine to give 1 mole of ethanedioate ions and 2 moles of yellow precipitate.

Deduce the structural formulae of compounds **L**, **M** and **N**.

[7]
Total: [20]

6 **(a)** **(i)** The Period 3 elements vary in their melting points, electrical conductivities and pH of aqueous solutions of their chlorides. Sketch a clearly labelled graph to illustrate the pH of solutions of the highest chloride of the Period 3 elements (from Na to P).

(ii) Describe what happens when aluminium chloride, $AlCl_3$, is dissolved in water, writing an equation to illustrate this.

(iii) Draw the structure of the compound formed when ammonia gas is pumped into a container with aluminium chloride.

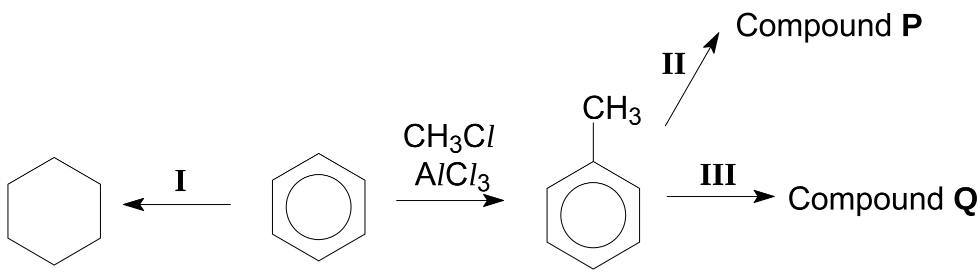
Hence, suggest the change in bond angle with respect to the aluminium atom in the structures before and after the reaction.

(iv) Boron trifluoride is also capable of similar reaction with ammonia. Explain why this is so.

(v) Boron trifluoride gas and aluminium chloride liquid are common catalysts used in isomerism, esterification and condensation reactions. Comment on the different physical states of the two catalysts.

[8]

(b) **(i)** The following reaction scheme shows some reactions involving benzene



Compound **Q** forms phenylethene in a 4-step synthesis reaction.

(i) State the type of reaction that took place for the conversion of methylbenzene from benzene.

(ii) State the type of reaction that took place in reaction **I**.

(iii) From the information given in the reaction scheme, draw the structures of compound **P** and **Q** and propose the reagents and conditions required for reaction **II** and **III**. Compounds **P** and **Q** are isomers.

[6]

(c) The ore, bauxite, consists of various forms of hydrated aluminium oxide, $Al_2O_3.nH_2O$. It is the raw material from which aluminium is obtained. The ore is usually found as a mixture of Al_2O_3 and Fe_2O_3.

Some data on the properties of Al_2O_3 and Fe_2O_3 is tabulated below.

	Al_2O_3	Fe_2O_3
Melting point (°C)	2072	1560
Solubility in hot concentrated sodium hydroxide	soluble	Insoluble
Molar mass (g mol^{-1})	101.96	159.69

(i) Suggest with the aid of an equation why aluminium oxide is soluble in hot concentrated sodium hydroxide whereas iron(III) oxide is not soluble.

(ii) Draw the dot-and-cross diagram for aluminium oxide.

(iii) With the aid of the *Data Booklet*, explain why aluminium oxide has a higher melting point as compared to iron(III) oxide.

(iv) Suggest a reason why BeO would exhibit similar chemical properties as Al_2O_3.

[6]
Total: [20]

END

BLANK PAGE

BLANK PAGE